

# PARUSHURAM HARIJANA

## Machine Learning Developer

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### Summary

Machine Learning Developer with strong skills in **Machine Learning, Computer Vision, Backend Development, and Data Structures and Algorithms**. Skilled in building production-ready ML systems and integrating models into scalable backend applications. Focused on solving complex problems and building efficient, scalable, and reliable software systems.

### Skills

Programming Languages: Python, Java, TypeScript, JavaScript

Frameworks and Libraries: PyTorch, NumPy, OpenCV, YOLOv5, Langchain, Node.js, Express.js, Fast API, Backend Development

Core ML Concepts: Computer Vision, Deep Learning, Object Detection, Object Tracking, LLMs, RAG, REST APIs, OOPS, DSA

Databases and Tools: MongoDB, SQL, Mongoose, Git, Docker, VS Code, Postman

### Experience

#### Cube Highways

Hyderabad, India

#### Machine Learning Developer

07/2024 - Present

- Developed and deployed **YOLOv5-based computer vision models** for large-scale object detection (highway roads assets) from real-time videos and image segmentation, optimizing accuracy and inference performance for production use.
- Designed and implemented end-to-end machine learning video inference pipelines, including preprocessing, post-processing, and result validation for real-world data.
- Built and maintained scalable backend services and **REST APIs** to map model detections to real-world entities, ensuring data consistency and reliable downstream consumption.
- Automated inspection and validation workflows by integrating machine learning outputs with backend systems, resulting in approximately **60% reduction in manual inspection effort**.

#### Cube Highways

Hyderabad, India

#### Machine Learning Intern

10/2023 - 06/2024

- Developed machine learning models for real-time object detection, gained hands-on experience with end-to-end ML workflows.
- Preprocessed and annotated large-scale video datasets to support effective model training and evaluation.
- Built **Node.js REST APIs** to store ML inference results in **MongoDB**, enabling seamless backend integration with detection pipelines.

### Education

#### Rajiv Gandhi of University of Knowledge Technologies

RK Valley, AP

Bachelor of Technology in Computer Science | GPA: **8.7** / 10

06/2020 - 04/2024

### Projects

#### Highway Road Assets and Cracks Detection

**Tech Stack** Python, OpenCV, PyTorch, YOLOv5, MongoDB, NodeJS, DeepSORT

- Developed a **real-time Computer Vision pipeline** to detect road assets (signboards, traffic lights, cracks) from highway video feeds.
- Designed end-to-end inference pipelines, including preprocessing and post-processing, to ensure consistent predictions.
- Integrated ML inference with backend services and databases, automating inspection workflows and achieving a ~60% reduction in manual inspection effort.

#### AI Assistant for Enterprise Analytics

**Tech Stack** Python, Langchain, LLM APIs, Vector Database(Chroma)

- Built a Text-to-SQL system that converts natural language questions into validated SQL queries for querying large internal databases.
- Implemented RAG-based grounding by combining live SQL query results with LLM reasoning to deliver accurate, human-readable analytics insights for non-technical users.